

Lecture #8: Joint Distributions

MATH 321 – Mathematical Statistics

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Inequalities

If we know a probability distribution (either the PDF or the CDF), we know, in theory, all the probabilities associated with it.

But what if we only know some summary statistics for X ?

We will use some theoretical inequality results to develop this idea.

Markov's Inequality

Theorem: Let $X \geq 0$ be a non-negative random variable. Then, for any $\varepsilon > 0$, we have

$$P[X \geq \varepsilon] \leq \frac{E[X]}{\varepsilon}.$$

This can also be written as

$$\varepsilon P(X \geq \varepsilon) \leq E[X].$$

Example: The number of items sold in a week is a random variable with mean 16. Estimate the probability that next week's sales exceed 25.

Tchebyshev's Inequality

Theorem: Let X be a random variable with mean μ . Then for any $\varepsilon > 0$, we have

$$P[|X - \mu| \geq \varepsilon] \leq \frac{V[X]}{\varepsilon^2}.$$

If we set $\varepsilon = k\sigma$ for some positive integer k , we get the more familiar form of Tchebyshev's Inequality:

$$P[|X - \mu| \geq k\sigma] \leq \frac{1}{k^2}.$$

Example: Let $X \sim \text{Uniform}(0, 2\sqrt{2})$. Find the bound of Chebyshev's inequality for the probability $P[|X - \mu| \geq 1]$.

Theorem: Let X be a random variable. The, for any $\varepsilon \geq 0$, we have that

$$P[X \geq \varepsilon] \leq e^{-\varphi(\varepsilon)},$$

where

$$\varphi(\varepsilon) = \max_{s>0} \left\{ s\varepsilon - \log M_X(s) \right\},$$

and $M_X(s)$ is the moment-generating function.

$\varphi(\varepsilon)$ is called the Fenchel-Legendre dual function of $\log M_X$.

Hoeffding's Inequality

Theorem: Let $X_1, X_2, \dots, X_N$ be i.i.d. random variables with $0 \leq X_i \leq 1$, and $E[X_n] = \mu$. Then

$$P\left[|\bar{X}_N - \mu| > \epsilon\right] \leq 2e^{-2\epsilon^2 N},$$

where $\bar{X}_N = \frac{1}{N} \sum_{i=1}^N X_i$.

Sample Averages

The **sample average** of a sequence of random variables $X_1, \dots, X_N$ is

$$\bar{X}_N = \frac{1}{N} \sum_{i=1}^N X_i.$$

This is a linear combination of random variables, so it too is a random variable.

Thus, it has an expectation, variance, PDF, CDF and everything.

Sample Averages

If $X_1, \dots, X_N$ are i.i.d., then the summary statistics $E(\overline{X}_N)$ and $V(\overline{X}_N)$ are easy.

But what is the *probability distribution* of $\overline{X}_N$?

The Weak Law of Large Numbers

Theorem: Let $X_1, \dots, X_N$ be a set of i.i.d. random variables with mean μ and variance σ^2 . Assume $E[X^2] < \infty$. Let $\bar{X}_N = \frac{1}{N} \sum_{i=1}^N$. Then, for any $\varepsilon > 0$,

$$\lim_{N \rightarrow \infty} P\left[|\bar{X}_n - \mu| > \varepsilon\right] = 0.$$

Convergence in Probability

A sequence of random variables $A_1, \dots, A_N$ **converges in probability** to a deterministic number α if for every $\varepsilon > 0$,

$$\lim_{N \rightarrow \infty} P\left[|A_N - \alpha| > \varepsilon\right] = 0.$$

We write $A_N \xrightarrow{p} \alpha$ to denote convergence in probability.

Example: Let $X_1, \dots, X_N$ be i.i.d. random variables with $X_i \sim \text{Uniform}(0, 1)$. Define $A_N = \min\{X_1, \dots, X_n\}$. Show that A_N converges in probability to zero.

Convergence

There are many types of convergence.

A sequence of random variables $A_1, \dots, A_N$ **converges almost surely** to α if

$$P\left[\lim_{N \rightarrow \infty} A_N = \alpha\right] = 1, \quad \text{or} \quad P\left[\lim_{N \rightarrow \infty} |A_N - \alpha| < \varepsilon\right] = 1.$$

We write $A_N \xrightarrow{a.s.} \alpha$ to denote almost sure convergence.

The Strong Law of Large Numbers

Theorem: Let $X_1, \dots, X_N$ be a sequence of i.i.d. random variables with common mean μ and variance σ^2 . Assume $E[X^4] < \infty$. Let

$\bar{X}_N = \frac{1}{N} \sum_{i=1}^N$ be the sample average. Then

$$P\left[\lim_{N \rightarrow \infty} \bar{X}_N = \mu\right] = 1.$$

The Central Limit Theorem

Theorem: Let $X_1, \dots, X_N$ be a sequence of i.i.d. random variables of mean $E[X_n] = \mu$ and variance $V[X_n] = \sigma^2$. Also assume that

$E[|X_n^3|] < \infty$. Let $\bar{X}_N = \frac{1}{N} \sum_{i=1}^N$ be the sample average and let

$Z_N = \sqrt{N} \left(\frac{\bar{X}_N - \mu}{\sigma} \right)$. Then

$$\lim_{N \rightarrow \infty} F_{\bar{Z}_N}(z) = F_Z(z),$$

where $Z \sim \text{Normal}(0, 1)$.

The Central Limit Theorem Examples

Examples:

- 1 Suppose we have 30 i.i.d. $X_i \stackrel{i.i.d.}{\sim} \text{Binomial}(n = 20, p = 0.25)$.
What is the probability that the sample mean of these 30 variables is above 8?

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What is the probability that the sample mean of these 30 variables is above 8?
- 2 Americans males is, on average, 69 inches tall, with a standard deviation of 3 inches. If I select a sample of 9 American males, what is the probability that our sample mean is within 0.5 of an inch?

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- 2 Americans males is, on average, 69 inches tall, with a standard deviation of 3 inches. If I select a sample of 9 American males, what is the probability that our sample mean is within 0.5 of an inch?
- 3 Suppose I want my sample mean to be within the 1 inch of the true population mean, with 95% probability. How many people do I need to include in my sample?